

## PRELIMINARY DATA: Assessment for Delta Operations on ESA and CESA-listed Salmonids

Last updated: *December 23, 2025*

For more detailed data on salmonid conditions in the Delta see corresponding webpage on [SacPAS](#).

### Executive Summary

- Entrainment management season is not active at this time.
- No salmonids have been salvaged this season.
- Winter-run presence in the delta is high.

### Natural Winter-run Chinook

#### Juvenile Production Estimate

The Juvenile Production Estimate for winter-run has not been established for the current water year.

### Current Status

*Entry Timing* - Historically, as of Dec 22, 59% of length-at-date (LAD) winter-run have entered the delta based on Knights Landing RST catch, 2% have exited the delta based on Chipps Island Trawl Catch, and 4% of DNA confirmed winter-run have been salvaged.

Table 1: Average Percent of annual emigrating population for unclipped LAD winter-run captured at the following locations and LAD and DNA confirmed winter-run salvaged at SWP and CVP Delta facilities for the past 10 years.

| Species                               | Red Bluff<br>Diversion<br>Dam | Tisdale<br>RST | Knights<br>Landing<br>RST | Sac Trawl<br>(Sher-<br>wood) | Chipps<br>Island<br>Trawl | Salvage |
|---------------------------------------|-------------------------------|----------------|---------------------------|------------------------------|---------------------------|---------|
| Chinook, LAD                          | 96%                           | 58%            | 59%                       | 34%                          | 2%                        | 4%      |
| Winter-run, Unclipped                 |                               |                |                           |                              |                           |         |
| Chinook, DNA                          | NA                            | NA             | NA                        | NA                           | NA                        | 4%      |
| Winter-run, Unclipped<br>(Water Year) |                               |                |                           |                              |                           |         |

*Red Bluff Diversion Dam Passage Estimate* - As of Dec 16 estimated passage to date of LAD winter run at Red Bluff Diversion is approximately 3.61 million fish. \* *Note that outmigration timing overlaps with spring run migrating fish, and true winter-run abundance likely differs from these estimates.*

*Delta Monitoring* - Total catch of LAD winter run at RSTs at Delta Entry (Tisdale, Knights Landing, Lower Sacramento River) between Dec 09 and Dec 20 is 21 individuals. Total catch at Sacramento Trawl and Beach Seines in the delta between Dec 10 and Dec 22 is 86 individuals. Total catch at Delta Exit at Chipps Island between Dec 10 and Dec 22 is 0 individuals.

## Annual Loss

The Juvenile Production Estimate has not been established for the current water year so thresholds are absent or are included from the previous water year. As of December 22 cumulative loss of genetically confirmed winter-run is 0 or 0.00% of the annual loss threshold. Cumulative loss in the past 7 days has been 0.

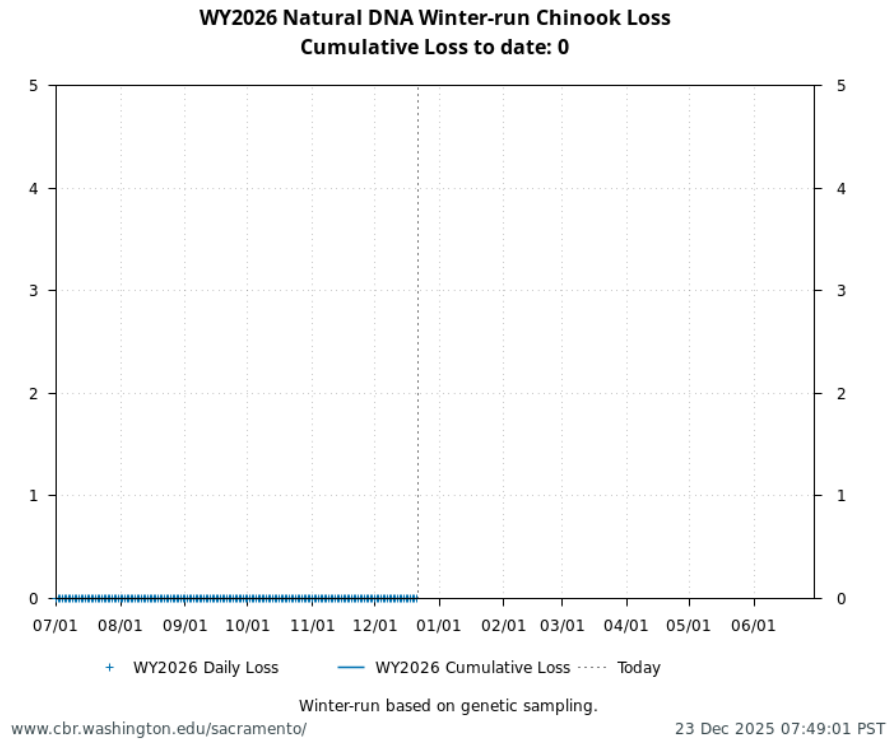


Figure 1: Cumulative loss of natural-origin winter-run for WY 2026. Cumulative loss is based on genetically confirmed winter-run captured in salvage or length-at-date winter-run in which genetic confirmation was unable to be obtained

## STARS

The Delta STARS Model is an individual-based simulation model that predicts survival, travel time, and routing of juvenile salmon migrating through the Sacramento–San Joaquin River Delta. This model gives insight into survival and routing patterns of winter-run based on most current conditions.

As of December 22, overall through delta STARS estimated survival probability (with 80% credible intervals) is 0.71 (0.67-0.74) placing it in the 97th percentile of historical STARS survival estimates for the month of December (WYs 2018-2025). STARS estimated routing and survival probabilities (with 80% credible intervals) into the interior delta are 0.11 (0.09-0.13) and 0.58 (0.48-0.67), respectively, corresponding to the 2nd and 98th percentiles of historical December estimates (WYs 2018-2025).

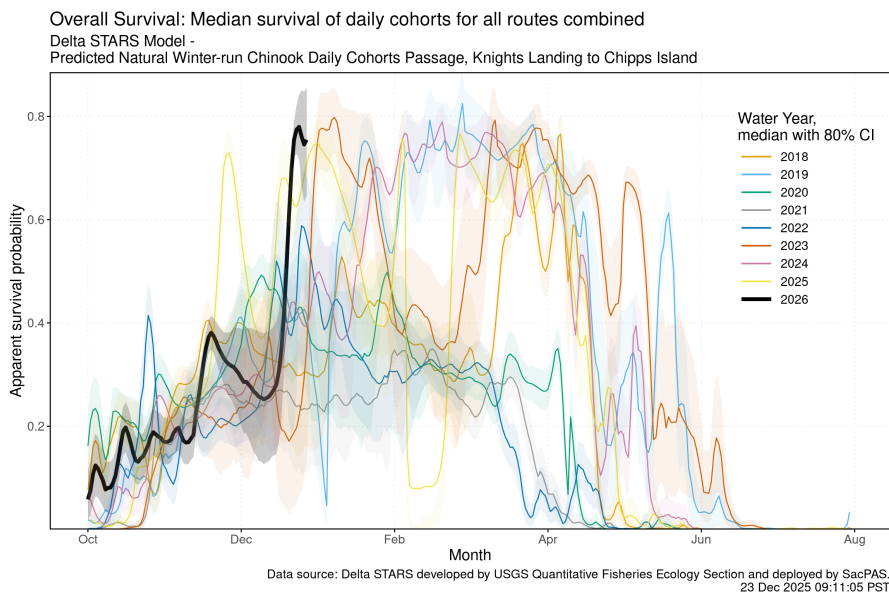


Figure 2: Estimated overall winter-run survival from Knights Landing to Chipps Island. Black line indicates the current water-year, and other colored lines correspond to past water years.

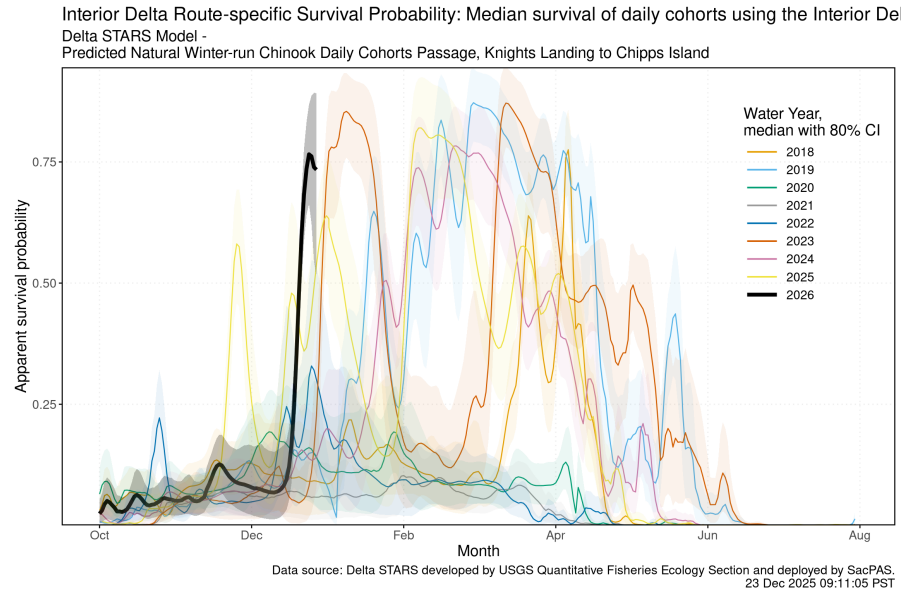


Figure 3: Estimated survival from Knights Landing to Chipps Island of simulate winter-run cohorts that route through the interior delta. Black line indicates the current water-year, and other colored lines correspond to past water years.

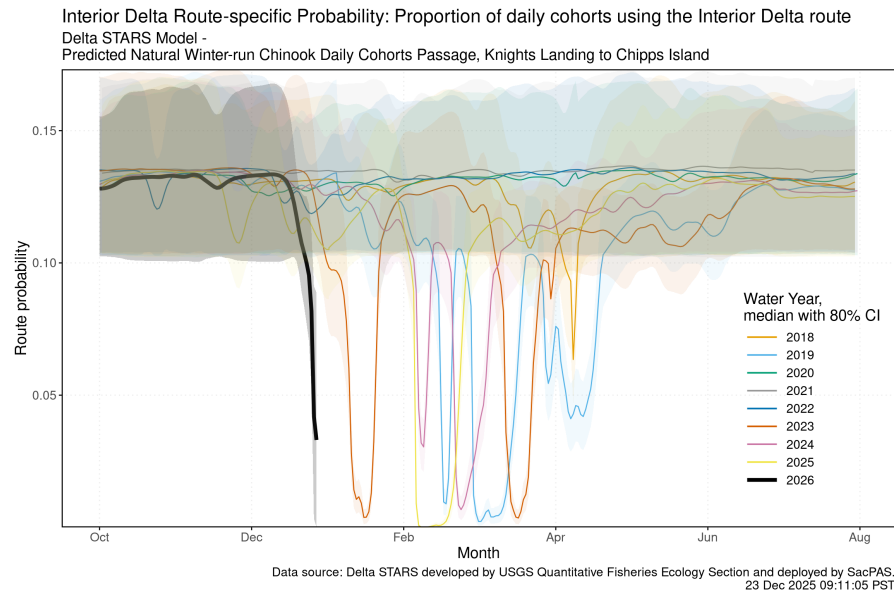


Figure 4: Estimated probability of winter-run routing into the interior delta. Black line indicates the current water-year, and other colored lines correspond to past water years.

## Hatchery Winter-run Chinook

### Hatchery Releases

To date, no winter-run Livingstone hatchery releases have occurred in WY 2026

### Juvenile Production Estimate

The Juvenile Production Estimate for Livingstone Stone hatchery winter-run has not been established for the current water year.

### Annual Loss

The Juvenile Production Estimate has not been established for the current water year so thresholds are absent or are included from the previous water year. To date, no loss has occurred as no hatchery winter-run have been released.

## Natural-origin Central Valley Steelhead

### Current Status

*Delta Entry Timing* - Historically, as of Dec 22, 0% of CCV steelhead have entered the delta based on Knights Landing RST catch, 0% have exited the delta based on Chipps Island Trawl Catch, and 1% have been salvaged.

Table 2: Average Percent of annual emigrating population for unclipped CCV steelhead captured at the following locations and salvaged at SWP and CVP Delta facilities for the past 10 years.

| Species                 | Red Bluff<br>Diversion Dam | Tisdale<br>RST | Knights<br>Landing<br>RST | Sac Trawl<br>(Sherwood) | Chipps<br>Island<br>Trawl | Salvage |
|-------------------------|----------------------------|----------------|---------------------------|-------------------------|---------------------------|---------|
| Steelhead,<br>Unclipped | NA                         | NA             | NA                        | NA                      | NA                        | 1%      |

*Delta Monitoring* - Total catch of unclipped steelhead at RSTs at Delta Entry (Tisdale, Knights Landing, Lower Sacramento River) between Dec 09 and Dec 20 is 0 individuals. Total catch at Sacramento Trawl and Beach Seines in the delta between Dec 10 and Dec 22 is 0 individuals. Total catch at Delta Exit at Chipps Island between Dec 10 and Dec 22 is 0 individuals.

## Annual Loss

As of December 22 cumulative loss of unclipped steelhead is 0 or 0.00% of the annual loss threshold. Cumulative loss in the past 7 days has been 0.

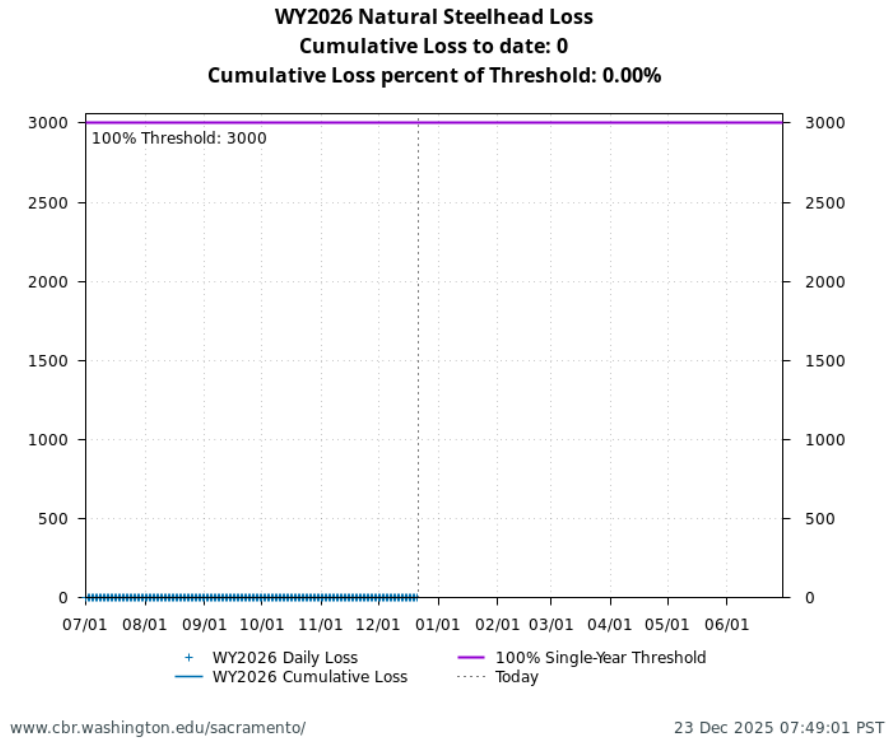


Figure 5: Cumulative loss of natural-origin and hatchery steelhead for WY 2026.

## Hatchery-origin Central Valley Steelhead

### Hatchery Releases

### Juvenile Production Estimate

### Spring-run

### Current Status

*Delta Entry Timing-* Historically, as of Dec 22, 16% of LAD spring run have entered the delta based on Knights Landing RST catch, 0% have exited the delta based on Chipps Island Trawl Catch, and 0% have been salvaged.

Table 3: Average Percent of annual emigrating population for LAD Spring-run Chinook Salmon captured at the following locations and salvaged at SWP and CVP Delta facilities for the past 10 years.

| Species                                  | Red Bluff<br>Diversion<br>Dam | Tisdale<br>RST | Knights<br>Landing<br>RST | Sac Trawl<br>(Sherwood) | Chippis<br>Island<br>Trawl | Salvage |
|------------------------------------------|-------------------------------|----------------|---------------------------|-------------------------|----------------------------|---------|
| Chinook, LAD<br>Spring-run,<br>Unclipped | 11%                           | 11%            | 16%                       | 3%                      | 0%                         | 0%      |

*Red Bluff Diversion Dam Passage Estimate* - As of Dec 16 estimated passage to date of LAD spring run at Red Bluff Diversion is approximately 0.01 million fish. \* *Note that outmigration timing overlaps with winter run and fall run outmigration, and true spring run abundance likely differs from these estimates.*

*Delta Monitoring*- Total catch of LAD spring run at RSTs at Delta Entry (Tisdale, Knights Landing, Lower Sacramento River) between Dec 09 and Dec 20 is 16 individuals. Total catch at Sacramento Trawl and Beach Seines in the delta between Dec 10 and Dec 22 is 84 individuals. Total catch at Delta Exit at Chippis Island between Dec 10 and Dec 22 is 0 individuals.

### Surrogate Releases

A total of 75,119 yearling surrogate fish have been released across 1 releases and a total of 0 young-of-year surrogate fish have been released across 0 releases. See details in table below.

Table 4: Spring-run Chinook Salmon Hatchery Surrogate Summary Table.

| Type     | Hatchery       | Release<br>Start | Release<br>End | Race          | # of CWT<br>Fish<br>Released | Loss Threshold (1%<br>of CWT Released) | Loss (% of<br>threshold) |
|----------|----------------|------------------|----------------|---------------|------------------------------|----------------------------------------|--------------------------|
| Yearling | Coleman<br>NFH | 2025-<br>11-17   | 2025-<br>11-17 | Late-<br>Fall | 75119                        | 751.19                                 | 0 (0.00%)                |

### Evaluation

1. **After January 1, has the first winter-run Chinook salvage occurred?** These criteria for the start of entrainment management for winter-run Chinook Salmon have not occurred.

## References